

Screenshots:

In Below screenshot, we can see successful searching operation, In- order traversal and deletion of leaf node.

```
Search for Tushar: 123-456-7890
Search for Khushi: 456-789-0123
Search for Rahul (not in tree): None
```

```
In-order Traversal:
Amol: 234-567-8901
Khushi: 456-789-0123
Pushpak: 567-890-1234
Tushar: 123-456-7890
Zakir: 345-678-9012
```

```
After Deleting Pushpak:
Amol: 234-567-8901
Khushi: 456-789-0123
Tushar: 123-456-7890
Zakir: 345-678-9012
```

In the screenshot below, we can see deletion operation of Node with One Child.

```
Search for Tushar: 123-456-7890
Search for Khushi: 456-789-0123
Search for Rahul : None
```

```
In-order Traversal:
Amol: 234-567-8901
Khushi: 456-789-0123
Pushpak: 567-890-1234
Tushar: 123-456-7890
Zakir: 345-678-9012
```

```
After Deleting Khushi:
Amol: 234-567-8901
Pushpak: 567-890-1234
Tushar: 123-456-7890
Zakir: 345-678-9012
```

In the screenshot below, we can see deletion operation of Node with Two Child.

```
Search for Tushar: 123-456-7890
Search for Khushi: 456-789-0123
Search for Rahul : None
```

```
In-order Traversal:
Amol: 234-567-8901
Khushi: 456-789-0123
Pushpak: 567-890-1234
Tushar: 123-456-7890
Zakir: 345-678-9012
```

```
After Deleting Tushar:
Amol: 234-567-8901
Khushi: 456-789-0123
Pushpak: 567-890-1234
Zakir: 345-678-9012
```

In the screenshot below, we can see deletion operation if name is not present.

```
Search for Tushar: 123-456-7890
Search for Khushi: 456-789-0123
Search for Rahul : None
```

```
In-order Traversal:
Amol: 234-567-8901
Khushi: 456-789-0123
Pushpak: 567-890-1234
Tushar: 123-456-7890
Zakir: 345-678-9012
```

```
After Deleting Rahul:
Amol: 234-567-8901
Khushi: 456-789-0123
Pushpak: 567-890-1234
Tushar: 123-456-7890
Zakir: 345-678-9012
```